Laura Gwilliams

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Education

2015–2020 *Ph.D., Psychology*
New York University, USA
Thesis Title: Towards a mechanistic account of speech comprehension

2012–2013 *M.Sc., Cognitive Neuroscience of Language*
Basque Center on Cognition, Brain and Language (BCBL), Spain

2009–2012 *B.A., Linguistics*
Cardiff University, UK

Research positions

2023-present *Assistant Professor*, Department of Psychology, Stanford University
Faculty Scholar Wu Tsai Neurosciences Institute and Stanford Data Science
PI of the Laboratory of Speech Neuroscience (GLySN) Lab
Co-director of The Center for Neural Data Science
Faculty Director of The Koret Human Neurosciences Community Laboratory

2020-2023 *Post-doctoral Fellow*, University of California, San Francisco

2013–2015 *Research Assistant*, New York University Abu Dhabi

Grants and Awards

2026 *Non-profit Philanthropy*, $6,917,250 (lead PI)

2025 *Big Ideas in Neuroscience*, Wu Tsai Neurosciences Institute, $1,000,000 (lead PI)

2025 *Neuroscience Fellowship Award*, Klingenstein Philanthropies, $450,000 (lead PI)

2025 *Transdisciplinary Initiatives Program*, Maternal and Child Health Research, $200,000 (co-PI)

2024 *Early Career Research Grant*, Whitehall Foundation, $300,000 (lead PI)

2024 *Community of Shared Research Platforms*, Stanford University, $1,982,000 (lead PI)

2023 *BRAIN Research Award*, The BRAIN Foundation, $178,202 (lead PI)

2022 *Trainee Professional Development Award*, Society for Neuroscience (SfN)

2021 *Glushko Dissertation Prize*, The Cognitive Science Society

2021 *Douglas H. and Katharine Fryer Thesis Award*, New York University

(Award for Best Doctoral Thesis)

2020 *Dissertation Award*, Society for the Neurobiology of Language

2020 *Martin Braine Fellowship*, New York University

2019 *William Orr Dingwall Dissertation Fellowship*

Fellowship in the Cognitive, Clinical, and Neural Foundations of Language

2019 *Facebook PhD Fellowship*, Facebook (Finalist)

2018 *Trainee Professional Development Award*, Society for Neuroscience (SfN)

2018 *Poster Prize*, Salzburg Mind Brain Annual Meeting (SAMBA)

2018 *Travel Award*, Cognitive Modelling and Computational Linguistics

2017 *Travel Award*, Cognitive Computational Neuroscience Conference

2016 *Travel Award*, Society for the Neurobiology of Language Conference

2012 *Dell Hymes Commendation for Academic Achievement*, Cardiff University

(Awarded to the top graduating student within the department)

Publications

Preprints

[1] de Heer Kloots, M., Kazemian, A., Turner, W., Parvizi, J. & **Gwilliams, L**. (submitted). Distinct yet neighboring neural populations encode past, future, and surrounding speech context in the human temporal lobe. bioRxiv

[2] McPherson, A., Sanchirico, S., Xu, A., Larson, E., Kaestner, M., Turner, W. F., **Gwilliams, L** & Taulu, S. (submitted). Noise Optimization of Basic Signal Component Extraction for Cryogenic and On-Scalp Magnetoencephalography. bioRxiv

[3] Gillis, M., Kries, J., Wouters, J., ***Gwilliams, L** & *Vandermosten, M.. (submitted). Neural Phoneme Processing in Children with and without Dyslexia. bioRxiv

Peer-reviewed articles

[4] **Gwilliams, L**. (2026). Time after time: neural codes of speech comprehension. *Trends in Cognitive Sciences*. DOI: 10.1016/j.tics.2026.09.003

[5] Roll, N., Kries, K., **Gwilliams, L** & Shain, C.. (2026). Artificial Aphasias in Lesioned Language Models. *NeurIPS Main Track*. arxiv

[6] Ergin, I., Kries, J., Gupta, S., Papworth Burrel, M & **Gwilliams, L**. (2026). Measuring Naturalistic Speech Comprehension in Real Time. *Behavior Research Methods*. DOI: 10.3758/s13428-026-02941-1

[7] Roll, N., Kries, K., Jin, F., Wang, C., Finley, AM., Sumner, M., Shain, C. & **Gwilliams, L**. (2026). The Text Aphasia Battery (TAB): A Clinically-Grounded Benchmark for Aphasia-Like Deficits in Language Models. *Proceedings of the 11th Workshop on Computational Linguistics and Clinical Psychology*. DOI: 10.18653/v1/2026.clpsych-1.27

[8] Turner, W., Hogendoorn, H. & **Gwilliams, L**. (2026). On the speed of conscious perception: how soon is now? *Commentary in Behavioral and Brain Sciences*. DOI: 10.1017/S0140525X25103051

[9] Kries, J., *Vandermosten, M. & ***Gwilliams, L**. (2026). The spatio-temporal dynamics of phonetic encoding in aging and aphasia. *Journal of Neuroscience*. DOI: 10.1523/JNEUROSCI.1001-25.2025

[10] Bhaya-Grossman, I., Leonard, M., Zhang, Y., **Gwilliams, L**., Johnson, K., Lu, J. & Chang, E. (2025). Shared and language-specific phonological processing in the human temporal lobe. *Nature*. DOI: 10.1038/s41586-025-09748-8

[11] Zhang, Y., Leonard, M., Bhaya-Grossman, I., **Gwilliams, L**., & Chang, E. (2025). Dynamics of auditory word form encoding in human speech cortex. *Neuron*. DOI: 10.1016/j.neuron.2025.10.011

[12] **Gwilliams, L**., Marantz, A., Poeppel, D. & King, JR. (2025). Hierarchical dynamic coding coordinates speech comprehension in the brain. *Proceedings of the National Academy of Sciences*. DOI: 10.1073/pnas.2422097122

[13] Abrams, E., Marantz, A., Krementsov, I. & **Gwilliams, L**. (2025). Dynamics of pitch perception in the auditory cortex. *Journal of Neuroscience*. DOI: 10.1523/JNEUROSCI.1111-24.2025

[14] Reilly, J., ... **Gwilliams, L** ... (2025). What we mean when we say semantic: Toward a multidisciplinary semantic glossary. *Psychonomic bulletin & review.* DOI: 10.3758/s13423-024-02556-7

[15] **Gwilliams, L**., Bhaya-Grossman, I., Zhang, Y., Scott, T., Harper, S., Levy., D (2025). Computational Architecture of Speech Comprehension in the Human Brain. *Annual Reviews*. DOI: 10.1146/annurev-linguistics-031120-111245

[16] Degano, G., Donhauser, P., **Gwilliams, L**. Merlo, P., & Golestani, N. (2024). Speech prosody enhances the neural processing of syntax. *Communications Biology*. DOI: 10.1038/s42003-024-06444-7

[17] Zuanazzi, A., Ripollés, P., Lin, WM., **Gwilliams, L**., *King, JR & *Poeppel, D (2024). Negation mitigates rather than inverts the neural representations of adjectives. *PLOS Biology*. DOI: 10.1371/journal.pbio.3002622

[18] ***Gwilliams, L**., *Leonard, M.K., Sellers, K.K., Chung, J.E., Dutta, B., & Chang, E.F. (2023). Large-scale single-neuron speech sound encoding across the depth of human cortex. *Nature*. DOI: 10.1038/s41586-023-06839-2

[19] **Gwilliams, L**., Flick, G., Marantz, A., Pylkkanen, L., Poeppel, D. & King, J.R. (2023). Introducing MEG-MASC a high-quality magneto-encephalography dataset for evaluating natural speech processing. *Nature Scientific Data*. DOI: 10.1038/s41597-023-02752-5

[20] **Gwilliams, L**., Marantz, A., Poeppel, D. & King, J.R. (2023). Top-down information shapes lexical processing when listening to continuous speech. *Language, Cognition and Neuroscience*. DOI: 10.1080/23273798.2023.2171072

[21] *Chung, J.E., *Sellers, K.K., Leonard, M.K., **Gwilliams, L**., Xu, D., Dougherty, M., Kharazia, V., Welkenhuysen. M., Dutta, B., Chang, E.F. (2022). High density single-unit human cortical recordings using the Neuropixels probe. *Neuron*. DOI: 10.1016/j.neuron.2022.05.007

[22] **Gwilliams, L**., King, JR., *Marantz, A. & *Poeppel, D. (2022). Neural dynamics of phoneme sequences: Position-invariant code for content and order. *Nature Communications*. DOI: 10.1038/s41467-022-34326-1

[23] Dikker, S., Mech, EM., **Gwilliams, L**., West, T., Dumas, G. & Federmeier, KD. (2022). Exploring age-related changes in inter-brain synchrony during verbal communication. *Psychology of Learning and Motivation*. DOI: 10.1016/bs.plm.2022.08.003

[24] Iemi, L., **Gwilliams, L**., Samaha, J., Auksztulewicz, R., Cycowicz, Y., King, JR., Thesen, T., Doyle, W., Devinsky, O., Schroeder, C.E., Melloni, L. & Haegens, S. (2021). Ongoing neural oscillations influence behavior and sensory representations by suppressing neuronal excitability. *NeuroImage*. DOI: 10.1016/j.neuroimage.2021.118746

[25] ***Gwilliams, L**., *Blanco-Elorrieta, E., Marantz, A. & Pylkkänen, L. (2021). Perceptual adaptation to accented speech: prefrontal cortex aids attunement in auditory cortices. *Nature Scientific Reports*. DOI: 10.1038/s41598-020-79640-0

[26] **Gwilliams, L**. & King, JR. (2020). Recurrent processes support a cascade of hierarchical decisions. *eLife*. DOI: 10.7554/eLife.56603

[27] Dikker, S., Assaneo, F., **Gwilliams, L**., Wang, L. & Kösem, A. (2020). MEG and Language: Using Magnetoencephalography to Study the Neural Basis of Language. *Neuroimaging Clinics of North America*. DOI: j.nic.2020.01.004

[28] **Gwilliams, L**. (2020). Hierarchical oscillators in speech comprehension: A commentary on Meyer, Sun & Martin. *Language, Cognition and Neuroscience*. DOI: 10.1080/23273798.2020.1740749

[29] **Gwilliams, L**. (2019). How the brain composes morphemes into meaning. *Philosophical Transactions of the Royal Society B*. DOI: 10.1098/rstb.2019.0311

[30] Stockall, L., Manouildiou, C., **Gwilliams, L**., Neophytou, K., & Marantz, A. (2019). Prefix Stripping Re-Re-Re-visited: MEG Evidence. *Frontiers in Psychology*. DOI: 10.3389/fpsyg.2019.01964

[31] **Gwilliams, L**., & Wallisch, P. (2019). Immediate ambiguity resolution in speech perception based on prior acoustic experience. PsyArXiv

[32] **Gwilliams, L**., Linzen, T., Poeppel, D., & Marantz, A. (2018). In spoken word recognition the future predicts the past. *Journal of Neuroscience*. DOI: 10.1523/JNEUROSCI.0065-18.2018

[33] **Gwilliams, L**., Poeppel, D., Marantz, A., & Linzen, T. (2018). Phonological (un)certainty weights lexical activation. In *Proceedings of the 8th Workshop on Cognitive Modeling and Computational Linguistics (CMCL 2018)* (pp. 29-34). arXiv

[34] **Gwilliams, L**. & Marantz, A. (2018). Morphological representations are extrapolated from morpho-syntactic rules. *Neuropsychologia*. DOI: 10.1016/j.neuropsychologia.2018.04.015

[35] Brodbeck, C., **Gwilliams, L**. & Pylkkänen, L. (2016). Language in context: MEG evidence for modality general and specific responses to reference resolution. *eNeuro*. DOI: 10.1523/ENEURO.0145-16.2016

[36] **Gwilliams, L**., & King, JR. (2017). Performance-optimized hierarchical models only partially predict neural responses during perceptual decision making. *NIPS workshop: Cognitively Informed Artificial Intelligence: Insights From Natural Intelligence* bioRxiv

[37] **Gwilliams, L**., Lewis, G. & Marantz, A. (2016). Functional characterisation of letter-specific responses in time, space and current polarity using magneto-encephalography. *NeuroImage*. DOI: 10.1016/j.neuroimage.2016.02.057

[38] Brodbeck, C., **Gwilliams, L**. & Pylkkänen, L. (2015). EEG can track the time course of reference resolution in small visual worlds. *Frontiers in Psychology*. DOI: 10.3389/fpsyg.2015.01787

[39] **Gwilliams, L**. & Marantz, A. (2015). Tracking non-linear prediction in a linear speech stream: Influence of morphological structure on spoken word recognition. *Brain and Language*. DOI: 10.1016/j.bandl.2015.04.006

[40] **Gwilliams, L**., Monahan, P., & Samuel, A. (2015). Sensitivity to morphological composition: Evidence from grammatical and lexical decision tasks. *Journal of Experimental Psychology: Language, Memory and Cognition*. DOI: 10.1037/xlm0000130

[41] **Gwilliams, L**. & Fontaine, L. (2015). Indeterminacy in process type classification. *Functions of Language*. DOI: 10.1186/s40554-015-0021-x

[42] Politzer-Ahles, S. & **Gwilliams, L**. (2015). Involvement of prefrontal cortex in scalar implicatures: Evidence from magnetoencephalography. *Language and Cognitive Neuroscience*. DOI: 10.1080/23273798.2015.1027235

Published Datasets, Corpora and Open Source Code

[1] *Lewis, G., *van Rijn, P., **Gwilliams, L.**, Larrouy-Maestri, P., Poeppel, D. & Ghitza, O. NyU-BU contextually controlled stories Corpus: NUBUC. DOI: 10.5281/zenodo.4075183

[2] **Gwilliams, L.**, Flick, G., Marantz, A., Pylkkanen, L., Poeppel, D. & King, J.R. (2023). Introducing MEG-MASC a high-quality magneto-encephalography dataset for evaluating natural speech processing. *Nature Scientific Data*. DOI: 10.1038/s41597-023-02752-5

[3] Waskom, M., Larson, E., Brodbeck, C., Gramfort, A., Burns, S ... **Gwilliams, L.**, King, JR., Liu, D. nipy/PySurfer:0.10.0. [Link]

[4] Larson, E., Gramfort, A., Engemann, DA., Leppakangas, J., Brodbeck, C ... **Gwilliams, L.**, ... mne-python-v1.2.0 [Link]

Book chapters

[1] Stockall, L. & **Gwilliams, L.** (2026). Distributed morphology and neurolinguistics. In *The Cambridge Handbook of Distributed Morphology*.

[2] **Gwilliams, L.** & Marantz, A. (2022). Neural processing of morphological structure in speech production, listening and reading. In *Current Issues in the Psychology of Language*.

[3] **Gwilliams, L.** & Davis, M.H. (2021). Extracting language content from speech sounds: The information theoretic approach. In *The Auditory Cognitive Neuroscience of Speech Perception*. Link

[4] King, JR., **Gwilliams, L.**, Holdgraf, C., Sassenhagen, J., Barachant, A., Engemann, D., Larson, E. & Gramfort, A. (2020). Encoding and Decoding Framework to Uncover the Algorithms of Cognition. In *The Cognitive Neurosciences*.

Presentations

Invited talks (last 5 years)

[1] *Keynote Speaker, AMLaP*. Nijmegen, The Netherlands. (2027, August).

[2] *Keynote Speaker, International Symposium on Auditory and Audiological Research (ISAAR)*. Nyborg, Denmark. (2027, August).

[3] *Keynote Speaker, International Symposium on Brain Function and Cognition*. (2027, February).

[4] *Colloquium Speaker, Indiana University*. IN, USA. (2026, October).

[5] *Colloquium Speaker, University of Arizona*. AZ, USA. (2026, October).

[6] *Invited Speaker, The Brain Conference on Neural Foundations of Communication: Integrative Approaches Across Species*. Heraklion, Greece. (2026, October).

[7] *Keynote Speaker, Cutting Gardens (EEG and MEG methods)*. Online. (2026, September).

[8] *Keynote Speaker, From Quantum Science to Brain Science, NSF*. VA, USA. (2026, September).

[9] *Invited Speaker, GRC Neural Mechansisms of Animal Communication*. ME, USA. (2026, June).

[10] *Colloquium Speaker, Yale University*. CT, USA. (2026, May).

[11] *Colloquium Speaker, University of Rochester*. NY, USA. (2026, April).

[12] *Speaker at 'Rising Star' Symposium, Cognitive Neuroscience Society*. Vancouver, Canada. (2026, March).

[13] *Keynote Speaker, Alpine Brain Imaging Meeting*. Champéry, Switzerland. (2026, January).

[14] *Invited Speaker, NeurIPS workshop: Data on the Mind and Brain*. CA, USA. (2025, December).

[15] *Colloquium Speaker, University of Michigan*. MI, USA. (2025, November).

[16] *Invited Speaker, ICON 2025*. Porto, Portugal. (2025, September).

[17] *Invited Speaker, ESCOP 2025*. Sheffield, UK. (2025, September).

[18] *Invited Speaker, CCN workshop 2025*. Amsterdam, Netherlands. (2025, August).

[19] *Invited Speaker, CogHear 2025*. University of Maryland. (2025, June).

[20] *Plenary address, American Psychological Association*. DC, USA. (2025, May).

[21] *Keynote Speaker, ICTEAP-5*. Waseda, Japan. (2025, April).

[22] *Colloquium Speaker, UC San Diego*. CA, USA. (2025, January).

[23] *Colloquium Speaker, USC, Center for Computational Language Sciences*. CA, USA. (2024, November).

[24] *Plenary address, Society for Language Development*. Boston, MA, USA. (2024, November).

[25] *NSF workshop, New horizons in language science*. Alexandria, VA, USA. (2024, May).

[26] *Colloquium Speaker, University of California, Santa Cruz*. Santa Cruz, CA, USA. (2024, April).

[27] *Symposium speaker, ARO - Association for Research in Otolaryngology.*. CA, USA. (2024, February).

[28] *Colloquium Speaker, McGovern Institute Special Seminar, MIT*. MA, USA. (2024, February).

[29] *Colloquium Speaker, Johns Hopkins University*. MD, USA. (2024, February).

[30] *Keynote Speaker, Annual Meeting on Phonology (AMP)*. Online. (2023, October).

[31] *Colloquium Speaker, UC Irvine*. CA, USA. (2023, October).

[32] *Keynote Speaker, Neurolinguistics in Sweden; Lund University*. Lund, Sweden. (2023, June).

[33] *Invited Speaker, CogHear Workshop*. Maryland, USA. (2023, June).

[34] *Colloquium Speaker, Cambridge University*. Cambridge, UK. (2023, February).

[35] *Colloquium Speaker, Queen Mary University London*. London, UK. (2023, February).

[36] *Invited Speaker, NeuroSpin*. Paris, France. (2022, December).

[37] *Meta AI and ENS*. Paris, France. (2022, May).

[38] *Special Talk Series, Max Planck Institute for Psycholinguistics*. Neurobiology of language: Key issues and ways forward II. (2022, March).

Teaching

2023-	*Instructor*, Stanford University
	Psych 1, Undergraduate
2023-	*Instructor*, Stanford University
	Data Science for Neuroscience Capstone, Undergraduate and Graduate
2023-	*Instructor*, Stanford University
	Language Neuroscience Seminar, Graduate and Undergraduate

Supervision

2026-	*Joyce Tang*, Lab Manager, Stanford University
2026-	*Ilina Bhaya-Grossman*, Postdoc, Stanford University
2026-	*Corentin Puffay*, Visiting Postdoc, Stanford University
2026-	*Xan McPherson*, Postdoc, Stanford University
2025-	*Nora Beier*, Postdoc, Stanford University (NIH K99)
2025-	*Ricky Rojas*, Undergraduate, Stanford University Statistics
2025-	*Takateru Yamakoshi*, PhD Student, Stanford University Computer Science
2025-	*Xirong Hu*, PhD Student, Stanford University
2024-	*Homa Vahidi*, MD Student, Stanford University Medical School
2024-	*Atlas Kazemian*, PhD Student, Stanford University
2024-	*Caroline Kaicher*, PhD Student, Stanford University
2024-	*William Turner*, Postdoc, Stanford University (AAA)
2023-	*Irmak Ergin*, PhD Student, Stanford University
2023-	*Jill Kries*, Postdoc, Stanford University
2023-2025	*Ellie Abrams*, PhD Student, New York University
2023-	*Marianne de Heer Kloots*, PhD Student, University of Amsterdam

Service

2026-2029	Elected Secretary	*Neurobiology of Language Society*
2024–2026	DEI Representative	*Cognitive Computational Neuroscience*
2022–2026	Program Committee	*Cognitive Computational Neuroscience*
2025	PC Chair	*Cognitive Computational Neuroscience*
2026	Dissertation Chair	Chaofei Fan, *Stanford University*
2026	Quals Committee	Samantha Gray, *Stanford University*
2025	Thesis Committee	Veronica Boyce, *Stanford University*
2025	Thesis Committee	Andrew Perley, *Stanford University*
2024	Thesis Committee	Linnea Evanson, *Ecole Normale Superieure*
2024	Thesis Committee	Ajay Subramanian, *Stanford University*
2024	Dissertation Chair	Jiayi Lu, *Stanford University*
2024	Dissertation Chair	Nay San, *Stanford University*
2024	Faculty founder	Bay Area Language Processing Interest Group (BAyLI)
2023	Thesis Committee	Vinay Raghavan, *Columbia University*
2023	Thesis Committee	Jill Kries, *KU Leuven*
2022	Thesis Committee	Juliett Millet, *Université de Paris*
2022	Thesis Committee	Théo Desbordes, *Meta AI & Neurospin*
Ad-hoc	Reviewer	*Science, Nature, Nature Neuroscience, Nature Communications, Nature Human Behaviour, PNAS, eLife, PLOS Biology, Journal of Neuroscience, NeuroImage, Human Brain Mapping, Cognition, Frontiers in Neuroscience, Glossa, Neurobiology of Language, Experimental Psychology, European Journal of Neuroscience, Mind Brain & Education, Cerebral Cortex, Psychonomic Bulletin & Review, Brain & Language, PLOS ONE, Cortex*
Ad-hoc	Reviewer	*National Science Foundation (USA)*